// Data Engineering Meetup · Nov 2026

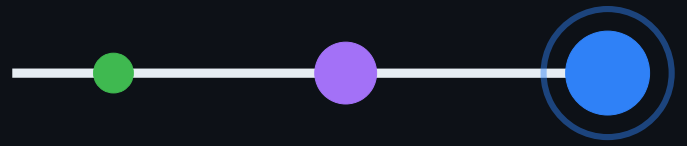
Metadata-Driven Pipeline Orchestration

From one YAML file to a running lakehouse



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// whoami

David Fengler

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I build data foundations — lakehouses, pipelines, governance — and turn them into AI that holds up in production, not just in demos.

- blog datadave.dev
- github github.com/phant0mw0lf
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- oss pipetree.dev

// agenda

What we'll cover

01 **The problem** why hand-built pipelines stop scaling

02 **The model** packages, layers and a YAML schema

03 **The executor** parallelism and failure handling

04 **Live demo** one config, one run

// 02

The model

Describe the pipeline, don’t write it



// 02 · the model

Configuration over code

One schema for every source, layer and target

Generated DAGs dependencies come from metadata, not from hand-wired tasks

Reviewable changes a new table is a pull request, not a notebook

// 02 · the model

Three layers, one contract

01

Raw

Land data as delivered. No business logic, full history.

02

Structured

Typed, deduplicated, conformed — still close to the source.

03

Curated

Modelled for consumers: reporting, ML features, agents.

// 02 · the model

A pipeline is just a file

pipelines/sales.yaml

```yaml
package: sales
layers:
  - name: raw
    source: jdbc://erp/orders
    schedule: "0 */2 * * *"   # every 2h
  - name: curated
    depends_on: [raw]
    retries: 3
```

// 03 · the executor

Before and after

hand-wired

- One notebook per table
- Retries copy-pasted
- Lineage lives in people’s heads

metadata-driven

- One engine, many configs
- Retries and timeouts declared once
- Lineage generated from the DAG

// 03 · the executor

What changed

1

engine instead of one notebook per table

0

hand-written dependency lists

n

tables onboarded by pull request

// 03 · the executor

Failures, made visible

Red node failed task

Grey nodes skipped downstream

Everything else keeps running



// takeaway

If you can describe it, you can generate it.

the one-line version of this talk

$ live demo_

pipetree run pipelines/sales.yaml

// thanks

Thank you.

Questions, war stories, pushback — all welcome.

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datadave.dev